# PRATIK KHANAL

WATER RESOURCES ENGINEER | HYDROINFORMATICS & EARTH OBSERVATION

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## RESEARCH PROFILE

Water resources engineer and AIT master's researcher connecting hydrodynamic modelling, satellite Earth observation and machine learning to flood and drought risk. Experience spans Himalayan flood hazards, citizen science monitoring and reproducible basin-scale decision support, with ongoing research on satellite-informed Noah-MP correction and flash drought in Thailand.

## EDUCATION

**Asian Institute of Technology, Thailand**
M.Eng., Water Engineering and Management | Aug 2025 - Present

GPA: 3.88/4.00 (first semester). GWSC and AIT scholarship. Coursework includes watershed hydrology, hydroinformatics, AI and big data, remote sensing/GIS, hydrodynamics and water resources systems.

**Himalaya College of Engineering, Tribhuvan University, Nepal**
B.E., Civil Engineering | Nov 2019 - Jun 2024

85.05% aggregate; Civil Engineering Faculty Topper; Tribhuvan University Third Rank. Four-year merit-based full scholarship. Final project: Dhap and Nagmati dam-break analysis, subsequently published in H2Open Journal.

## CURRENT & SELECTED RESEARCH

**Satellite-informed land surface reconstruction**
Ongoing research | AIT

Developing satellite-informed correction of Noah-MP land surface model outputs for surface and root-zone soil moisture, terrestrial water storage, precipitation and evaporation. The research targets a 1 km gridded record for 1940-2025 and its application to flash drought in Thailand. Satellite observations inform correction over their available overlap; the full historical period is a model-based reconstruction.

**Thailand Water Storage Observatory**
Research dashboard

Built basin-scale decision support for Thailand's 22 principal basins and national aggregate, integrating GRACE water storage, full-compartment groundwater estimates, climate reconstruction and observed groundwater comparisons. Includes 46 regional/target ensembles, independent temporal testing and evidence from 511 wells. Residuals retain model and observation uncertainty and do not establish pumping volumes.

**Bhotekoshi flood study, Nepal**
Project experience

Worked on flooding in the Bhotekoshi basin, Nepal, as part of a broader focus on Himalayan flood hazards and water resources. Detailed methods and project results are not yet included in the public portfolio.

## PROFESSIONAL EXPERIENCE

**Asian Institute of Technology**
Research Assistant | URBAN Project | Dec 2025 | Present

Flood hazard mapping under Dr. Natthachet Tangdamrongsub for the ADPC-funded URBAN project; HEC-RAS and GIS analysis of urban flood exposure, with forecast and vulnerability integration.

**Addon Engineering Solutions**
Hydrologist | Nov 2024 | Present

Hydrological and hydraulic analysis for flood protection in Kaski, Arghakhanchi and Gulmi; rainfall-runoff modelling, river hydraulics, sediment assessment and erosion-control design.

**Himalaya College of Engineering**
Teaching Assistant | Part-time | Nov 2024 | Apr 2025

Taught computer-aided civil drawings and supported surveying laboratories and field camps, including auto level and total station instruction.

**SmartPhones 4 Water Nepal**
Research Assistant | Jun 2024 | Nov 2024

Co-led MESSHI sediment monitoring with S4W-Nepal and the University of Pittsburgh; contributed to VICTORY precipitation and water-level quality control, citizen science monitoring and Hanumante flood modelling.

## SELECTED COMPUTATIONAL PROJECTS

**Nan River basin flood forecasting, Thailand**
Ongoing thesis research | AIT

Developing an integrated deep learning and hydraulic modelling framework for the Nan River basin. The proposed chain couples LSTM temporal learning with directed graph neural network routing for seven-day discharge forecasts, HEC-RAS 2D reference simulations and a ConvLSTM surrogate for flood depth, velocity and inundation extent. Planned evaluation uses RID discharge, TMD meteorology and Sentinel-1 flood observations, with a daily-updatable web decision-support prototype. Model skill and operational performance remain to be established.

**HydroEVT | Nepal Hydrological Extreme Flow DSS**
Collaborative research software | Pratik & Saugat

Co-developed HydroEVT and its Streamlit decision-support interface for Nepalese flood and low-flow frequency analysis. The workflow combines observed-record quality checks, stationarity and independence diagnostics, L-moment distribution fitting, goodness-of-fit ranking, return-period design flows and bootstrap uncertainty. Interactive station and catchment views communicate record quality and uncertainty alongside the estimates.

**Madi River streamflow forecasting**
Academic project | AIT, 2025

Developed and evaluated an LSTM model for multi-step streamflow forecasting, with hydro-meteorological time-series preparation and comparison against conventional forecasting methods.

**Terai multi-source drought analysis**
Academic project | AIT, 2025

Analysed soil moisture, precipitation and terrestrial water storage indicators using SMAP, ERA5 and GRACE, with Python processing and drought severity mapping for Nepal's Terai region.

## ADDITIONAL RESEARCH & SOFTWARE

**GeoAI landslide prediction & infrastructure risk**
Individual term report | AIT | April 2026

Prepared a term report on landslide susceptibility and building exposure in Bagmati Province, Nepal. Compared SE-DenseNet, TabTransformer and XGBoost using a landslide inventory and 13 environmental factors at 30 m resolution; combined susceptibility maps through spatial ensemble analysis and used SAM-based building detection in high-risk areas. Included case studies of the 2014 Jure and 2024 Jhyaple Khola landslides.

**Water quality & groundwater modelling reports**
Group term projects | AIT | April 2026

Co-authored QUAL2K and MODFLOW group reports with Mohammed Yameen and Farzeen Khan. The QUAL2K project examined dissolved oxygen, CBOD loading and interventions including weirs and riparian shading. The MODFLOW project compared steady-state groundwater heads, water balances and wetland discharge under alternative boundary conditions and pumping scenarios.

**Forest fire prediction**
Public code project

Compared multiple linear regression, Ridge, Lasso and Elastic Net for fire prediction in the Bezia region, using cross-validation and hyperparameter tuning to address model selection and overfitting.

## PEER-REVIEWED PUBLICATIONS

**Dam break analysis of the Nagmati and Dhap dams using HEC-RAS**
Khanal, P., et al. | H2Open Journal 8(3), 139-156 | 2025

DOI: 10.2166/h2oj.2025.058

**Comparative Analysis of Environmental Protection Provisions in Construction Procurement: A Case Study of Nepal's Projects Funded by National and International Agencies**
Acharya, S., et al. | International Journal on Engineering Technology 1(2), 250-267 | 2024

DOI: 10.3126/injet.v1i2.66695

**Evaluating the Alignment between Intended Project Durations and Standard Time Cost Models: A Comparative Analysis in Basic Hospital Construction Projects in Nepal**
Dahal, D., et al. | International Journal on Engineering Technology 1(2), 211-229 | 2024

DOI: 10.3126/injet.v1i2.66688

## TECHNICAL EXPERTISE

**Hydrology & hydraulics**

HEC-RAS, HEC-HMS, SWAT, MIKE, HEC-ResSim, HBV, QUAL2K, MODFLOW

**Earth observation & GIS**

GRACE, SMAP, ERA5, Noah-MP, ArcGIS, QGIS

**Scientific computing & AI**

Python, R, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras, LSTM, CNN, SAM

**Engineering & fieldwork**

AutoCAD Civil 3D, SW DTM, total station, auto level, acoustic Doppler velocimetry, Open OBS sediment sensing

## PROFESSIONAL REGISTRATION & ENGAGEMENT

Registered Civil Engineer, Nepal Engineering Council, No. 86044, Civil A category (Oct 2024).

Presenter, Kantipur Engineering College Conference 2024. Attendee, FIDIC Asia Pacific Conference 2024: Governance for Climate Shocks. Member, Himalaya Civil Club.

## ACADEMIC REFEREES

Dr. Natthachet Tangdamrongsub, Assistant Professor, Water Engineering and Management, AIT | natthachet@ait.ac.th

Prof. Sangam Shrestha, Water Engineering and Management, AIT | sangam@ait.ac.th